mid-term exam paper

course name __database principle__ examination time __120minutes__

student number________________ student name________________

1. Please briefly answer the following questions: (5+5+5+5=20)
   a) Explain the difference between files and Databases;
   b) Explain the difference between Data, Data model and Data Schema;
   c) Explain the four parts of SQL;
   d) How does the modern database system realize the independence of data?

2. The following relational modes are given: R = (A, B, C), S = (D, E, F). Let the relations r(R) and s(S) be known. Please give the tuple relational calculus expression equivalent to the following expression: (4+5+6=15)
   a) $\prod_{A}(r)$
   b) $\prod_{A,F}(\sigma_{C=D}(r \times s))$
   c) $r * \bowtie_{C=D} s$

3. Please draw the E-R diagram of the project relationship, the entities including employee, project, supplier, component. The relationship between the entities are as follows: (10)
   a) A project has multiple employees, and one employee can only work in one project;
   b) A project has multiple suppliers, and one supplier can supply multiple projects;
   c) A project requires multiple components, which can come from the same supplier or from different suppliers, and one kind of components can be used by multiple projects.

   Note: The entries have been given, please draw the E-R diagram that correctly represents these entities' relationships.

project	employee	supplier	component

4. Using SQL to finish following questions(6+6+10+11+12=45)

R1

sid	bid	day
22	101	10/10/96
58	103	11/12/96

B1

bid	bname	color
101	tiger	red
103	lion	green
105	hero	blue

S1

sid	sname	rating	age
22	dustin	7	45.0
31	lubber	8	55.5
58	rusty	10	35.0

S2

sid	sname	rating	age
28	yuppy	9	35.0
31	lubber	8	55.5
44	guppy	5	35.0
58	rusty	10	35.0

a) How to find name of the oldest sailor who reserved "tiger" boat?

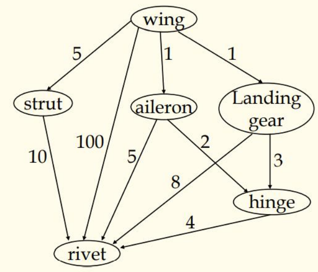
b) How to find names of sailors who have reserved boat #103 and reserved only one time?

c) How to find the name of boats which have been reserved more than two times. (Note: you should use operator: COUNT)

- A classical "parts searching problem"

Components

Part	Subpart	QTY
wing	strut	5
wing	aileron	1
wing	landing gear	1
wing	rivet	100
strut	rivet	10
aileron	hinge	2
aileron	rivet	5
landing gear	hinge	3
landing gear	rivet	8
hinge	rivet	4



Directed acyclic graph, which assures the recursion can be stopped

d) Find all subparts and their total quantity needed to assemble a wing.

e) Find the number of rivets used in each subpart (except rivets).

5. Answer the following questions: (4+6=10)

a) There is a borrow table and the data schema is as follows:

Borrow(studentID, bookID, date)

Point out the primary key of the borrow table.

b) There is a work relationship table and the data schema is as follows:

Work(warehouseID, employeeID, componentID, quantity)

One warehouse has multiple employees, and each employee only works in one warehouse. In each warehouse, one type of components is composed of one employee is responsible, but one employee can be responsible for a variety of components. Analyze all possible candidate keys for this pattern.

1. Please briefly answer the following questions:

a) Explain the difference between files and Databases;

Files vs. Databases

- Application must stage large datasets between main memory and secondary storage (e.g., buffering, page-oriented access, 32-bit addressing, etc.)
- Special code for different queries
- Must protect data from inconsistency due to multiple concurrent users
- Crash recovery
- Security and access control

b) Explain the difference between Data, Data model and Data Schema;

Data, Data Model and Data Schema

- *Data* are symbols for describing the things of real world. They are existing form of information.
- A *data model* is a collection of concepts and definitions for describing data.
- A *schema* is a description of a particular collection of data, using a given data model.
- The *relational model of data* is the most widely used model today.
  - Main concept: *relation*, basically a table with rows and columns.
  - Every relation has a *schema*, which describes the columns, or fields.

c) Explain the four parts of SQL;

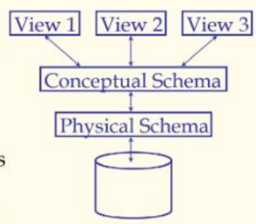
SQL Language

- It can be divided into four parts according to functions.
  - Data Definition Language (DDL), used to define, delete, or alter data schema.
  - Query Language (QL), used to retrieve data
  - Data Manipulation Language (DML), used to insert, delete, or update data.
  - Data Control Language (DCL), used to control user's access authority to data.
- QL and DML are introduced in detail in this chapter.

d) How does the modern database system realize the independence of data?

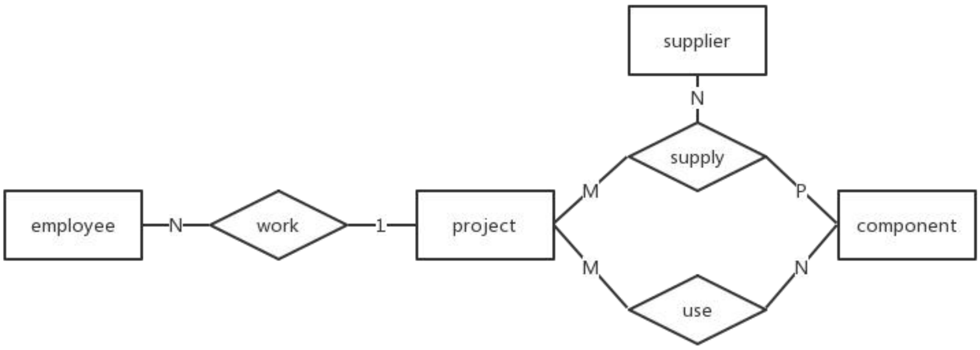
Levels of Abstraction: ANSI-SPARC Architecture

- Many *views*, single *conceptual (logical) schema* and *physical schema*.
  - Views describe how users see the data.
  - Conceptual schema defines logical structure
  - Physical schema describes the files and indexes used.



2. The following relational modes are given: R = (A, B, C), S = (D, E, F). Let the relations r(R) and s(S) be known. Please give the tuple relational calculus expression equivalent to the following expression:
   a) $\prod_A (r) = \{t(A) | t \in r\}$
   b) $\prod_{A,F} (\sigma_{C=D}(r \times s)) = \{t[AF] | t[ABC] \in r \wedge t[DEF] \in s \wedge t.C = t.D\}$
   c) $r * \bowtie_{C=D} s = \{t[ABCDEF] | (t[ABC] \in r \wedge t[DEF] \in s \wedge t.C = t.D) \vee (t[ABC] \in r \wedge \neg(t.C \in s[D]) \wedge t.D = NULL \wedge t.F = NULL \wedge t.F = NULL)\}$
3. Please draw the E-R diagram of the engineering relationship, the entities including employees, engineering, suppliers, components, the relationship between the entities are as follows:
   a) A project has multiple employees, and one employee can only work in one project;
   b) A project has multiple suppliers, and one supplier can supply multiple projects;
   c) A project requires multiple components, which can come from the same supplier or from different suppliers, and one kind of components can be used by multiple projects.

   Note: The entries have been given, please draw the E-R diagram that correctly represents these entities' relationships.



4. Using SQL to finish following questions

R1

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sid	sname	rating	age
28	yuppy	9	35.0
31	lubber	8	55.5
44	guppy	5	35.0
58	rusty	10	35.0

a) How to find name of the oldest sailor who reserved “tiger” boat?

```
SELECT sname
FROM (SELECT B.bid, max(S.age) AS m FROM Sailors AS S, Reserves AS
R, Boats AS B
WHERE S.sid = R.sid
AND R.bid = B.bid
AND B.bname = "Tiger"
GROUP BY B.bid) AS A, Sailors AS S2, Reserves AS R2
WHERE A.bid = R2.bid and S2.age = A.m and S2.sid = R2.sid;
```

b) How to find names of sailors who have reserved boat #103 and reserved only one time?

```
SELECT S.sname
FROM Sailors AS S
WHERE exists
(Select *
from Reserves R1
where R1.bid =103 AND R1.sid = S.sid AND
NOT exists
(select *
from Reserves R2
where R2.bid = 103 AND R2.sid = R1.sid AND R1.day <> R2.day) );
```

c) How to find the name of boats which have been reserved more than two times. (Note: you should use operator: COUNT)

```
SELECT bname
FROM Boats
WHERE bid in
(select bid
from Reserves
group by bid
having count(*)>2);
```

- A classical “parts searching problem”

Components

Part	Subpart	QTY
wing	strut	5
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wing	landing gear	1
wing	rivet	100
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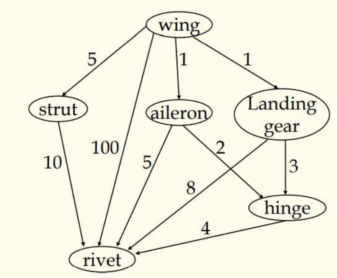

Directed acyclic graph, which assures the recursion can be stopped

d) Find all subparts and their total quantity needed to assemble a wing.

```
WITH wingpart (subpart, qty) AS
((SELECT subpart, qty
FROM components
WHERE part=‘wing’)
UNION ALL
(SELECT c.subpart, w.qty*c.qty
FROM wingpart w, components c
WHERE w.subpart=c.part))
SELECT subpart, sum(qty) AS qty
FROM wingpart
Group BY subpart ;
```

e) Find the number of rivets used in each subpart (except rivets).

```
WITH wingpart (part, qty) AS
((SELECT part, qty
FROM components
WHERE subpart=‘rivet’)
UNION ALL
(SELECT c.part, w.qty*c.qty
FROM wingpart w, components c
WHERE w.part=c.subpart))
SELECT part, sum(qty) AS qty
FROM wingpart
Group BY part ;
```

5. Answer the following questions:

e) There is a borrow table and the data schema is as follows:

   Borrow(studentID, bookID, date)

   Point out the primary key of the borrow table.

   All(reason omit)

f) There is a work relationship table and the data schema is as follows:

   Work(warehouseID, employeeID, componentID, quantity)

   One warehouse has multiple employees, and each employee only works in one warehouse. In each warehouse, one type of components is composed of one employee is responsible, but one employee can be responsible for a variety of components. Analyze all possible candidate keys for this pattern.

   a. {employeeID, componentID}
   b. {warehouseID, componentID}